

SOC Lab with Wazuh + Sysmon

□ Overview

This project demonstrates a complete SOC (Security Operations Center) lab built using VirtualBox, Wazuh SIEM, Sysmon, and Kali Linux.

□ Architecture

```
Attacker (192.168.100.30)
    ↓
Windows (192.168.100.20)
    ↓ logs
SIEM (192.168.100.10)
```

⚙️ □ Technologies

- VirtualBox
- Ubuntu 24.04
- Wazuh 4.12
- Windows 11
- Sysmon
- Kali Linux
- Nmap

□ Setup Steps

1. Create Network

- Host-Only Adapter
- Network: 192.168.100.0/24

2. Create VMs

- SIEM (Ubuntu)
- Windows 11
- Kali Linux

3. Configure IPs

- SIEM: 192.168.100.10
- Windows: 192.168.100.20
- Attacker: 192.168.100.30

4. Install Wazuh

```
curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh
bash wazuh-install.sh -a
```

5. Install Sysmon

```
sysmon64.exe -i sysmonconfig.xml
```

6. Connect Agent

- Add agent in Wazuh dashboard
- Install on Windows
- Start service

☐ Testing

- Ping between machines
- Run Nmap scans
- Verify logs in Wazuh dashboard

☐ Troubleshooting

- Fix NAT interface issues (enp0s3 vs eth0)
- Enable ICMP in Windows Firewall
- Verify IP forwarding

☐ Results

- Logs successfully ingested
- Vulnerability detected (CVE-2023-47359)
- Fully operational SOC pipeline

☐ Learning Outcomes

- SIEM deployment
- Network troubleshooting
- Log analysis
- Vulnerability management

☐ Future Improvements

- Attack simulations
- Custom detection rules
- Threat intelligence integration